AgriSentinel

Monetizing Fresh Satellite Substrate as Enforcement-Grade Dossiers for Illegal Farmland Factories

Big Data Systems — Final Project · National Taiwan University · Spring 2026

GitHub: github.com/HaoWen46/AgriSentinel **Live demo:** (optional bonus — insert deployed URL)
Reproduce with `make demo` (live) or `make demo-offline` (no network / no API key), then open `localhost:8000`.

Thesis. Raw, public, easy data has no durable value: in a competitive market its price falls to the marginal cost of production, and a 200-line scraper has a marginal cost near zero. The durable value is in **cognitive labour** — the work a human analyst does to turn messy inputs into a decision-grade artifact. As of 2026, AI agents perform that labour at a fraction of the human wage. **AgriSentinel sells the agent's output artifact — a per-parcel enforcement dossier — not the data underneath.** The big-data pipeline (multi-temporal satellite imagery fused with cadastral/zoning data and statute retrieval) exists to feed the agent a substrate fresh and well-structured enough that its dossier beats anything the customer could get from a generic chatbot. The exploitable asymmetry: **government has authority but no eyes.** It can lawfully cut water/electricity and demolish illegal farmland structures, but it does not continuously watch the farmland, so it learns of violations late. The technical edge is **collapsing the observation lag** — and there is no luck in it: either the pipeline sees the new structure or it doesn't.

1. Target Customer

Primary customer (the whale): county / municipal enforcement units. In Taiwan, farmland-factory enforcement sits with 縣市政府 經濟發展局 / 都市發展局 / 環境保護局. They hold the **authority** (工廠管理輔導法, 區域計畫法) and the **budget**, but lack continuous monitoring. Their process today is reactive: a citizen reports, an inspector is dispatched, paperwork (公文) is hand-drafted, and only then can a cut-off / demolition order proceed.

Beachhead / channel (moves faster): environmental NGOs already in the workflow — 地球公民基金會 (Citizens of the Earth Taiwan) and the g0v community behind **Disfactory** (disfactory.tw). Citizens report suspected factories, and CET files the paperwork that pressures local governments. They even crowdsource **detection** through a game, 大家來找廠, where volunteers eyeball aerial imagery to spot new buildings on farmland. That manual visual change-detection, plus manual paperwork drafting, is exactly the cognitive labour AgriSentinel automates — we do not invent demand, we replace a crowdsourcing game and hand-written dossiers with a pipeline, riding the NGO→government relationship as the cold-start wedge.

Why us over the status quo. The status quo is anonymous citizen reports + a volunteer game + manual paperwork: slow, unsystematic, and unscalable across the ≈50,000 sites. We deliver systematic, island-scalable detection **plus** a ready-to-file dossier per parcel — the same output an inspector would assemble by hand, produced in seconds.

2. Evidence of Demand and Willingness to Pay

All collection scripts are in `scripts/demand/` and are reproducible.

1 — The problem is real and growing (reproducible). `disfactory_stats.py` pulls reported suspected-factory records from the Disfactory public API around the pilot AOI and buckets them by report year. A representative run (7 km radius around 和美鎮, retrieved 2026-06) returned **100 reports within 12 km of one township alone**, sustained year after year:

Year	2020	2021	2022	2023	2024	2025	2026*
Reports near AOI	6	14	28	14	21	10	7

Cross-checked against public island-wide figures: ≈50,000 illegal factories on Taiwan's farmland, 3,000–6,000 new per year, ≈1,500 ha of farmland lost per year (CET reporting; Disfactory; MOEA statistics — re-verify and cite at submission). *2026 partial.

2 — The manual-labour baseline we replace. The cognitive labour has two halves, both currently human: (a) **detection** — the 大家來找廠 volunteer game, ≈1–2 minutes of careful review per image across tens of thousands of parcels per cycle; and (b) **dossier drafting** — hand-assembling the 地號, zoning status, evidence, the specific statute, and a recommended action into filing-ready paperwork. AgriSentinel automates (a) outright and reduces (b) to **human review of a pre-drafted dossier**. The survey instrument (`scripts/demand/survey_questions.md`) quantifies both: time-per-case for staff and minutes-per-image for volunteers (non-monetary WTP).

3 — Published willingness to pay (reproducible). Government WTP is **public record**. `tender_search.py` searches the e-procurement portal (政府電子採購網) via the g0v PCC API for monitoring / remote-sensing / illegal-use keywords. A representative run returned **370 matching tenders**:

Keyword	違章工廠	農地違規	航遙測	遙測影像	衛星影像	國土監測	無人機監測
Matches	5	21	100 †	100 †	100 †	10	31

† page-capped at 100 — true counts are higher. Government bodies already procure aerial / remote-sensing and land-monitoring services at scale; we attach to an existing budget line, not a hypothetical one.

4 — Commercial comparable (price anchor). A Japanese commercial service sells satellite + AI farmland monitoring to municipalities at ≈¥2,000,000 per municipality (≈NT$430,000), claiming ≈80% agreement with physical patrols and ≈3 weeks to deploy — proof the exact pipeline ships commercially.

5 — The defensible claim. N flagged parcels at precision P (measured against Disfactory labels, § 4); the incumbent is manual crowdsourcing + hand-written dossiers; comparable monitoring sells at ≈NT$430k/municipality; government already procures remote-sensing monitoring (370+ tenders). At a cost-per-parcel in cents (§ 5), the unit economics close at a handful of municipalities.

3. System Design

```
        SOURCES                  INGESTION                 LAKE / STORE
 Sentinel-2 L2A (STAC) ---> stac_fetch  (bi-temporal) ---> MinIO (S3): COGs, manifests
 Disfactory open API   ---> labels_fetch (eval set)   ---> PostGIS: parcels, labels,
 NLSC land-use/parcels ---> parcels_load                    detections, flagged, dossiers
 全國法規資料庫 statutes ---> laws_fetch -> chunk -> embed --> pgvector: law chunks
                                   |
                                   v
 PROCESSING:  change_detect (NDBI/NDVI; torchgeo-swappable)
              spark_tiles  (PySpark tiles the AOI, CD per tile)   -- batch -->
              zoning_join  (detections x farmland parcels, PostGIS ST_Intersects)
              evaluate     (precision / recall vs Disfactory labels)
                                   |
                                   v
 AGENT (value capture): dossier_agent
   RAG (pgvector) retrieves matching statute chunks + structured detection facts
   -> Claude drafts a grounded enforcement dossier -> adversarial draft->critic
                                   |
                                   v
 DELIVERY: FastAPI + Leaflet dashboard (flagged parcels, before/after imagery,
   downloadable dossiers, eval metrics). Kafka event: new detection -> worker
   -> dossiers -> live dashboard update (SSE).
```

Storage and processing — why each tool fits the shape of the data.

Technology	Justification
Sentinel-2 via Planetary Computer STAC	Free, no-auth, time-series enables change detection; Cloud-Optimized GeoTIFF range reads pull only the AOI footprint (≈0.6 MP/band) of a ≈110 MP scene — keeps volume tractable.
MinIO (S3) + COG	Raw lake: immutable raster zone, lakehouse pattern; variety + volume.
PostGIS + pgvector	One store, two jobs: GIST-indexed spatial joins decide is this farmland? (`ST_Intersects`), and a `vector` column serves the statute RAG (<=> cosine).
PySpark (`spark_tiles`)	Batch paradigm: tiles the AOI and maps change detection per tile; scales to many tiles / historical date pairs by adding executors. Single-process fallback included.
Apache Kafka (KRaft)	Velocity: a `detection.completed` event fans out to a worker (join→eval→dossiers) and to the dashboard over SSE — new imagery flows to fresh dossiers with no manual rerun. Any Kafka-API broker (e.g. Redpanda) works.
Anthropic Claude	The cognitive worker producing the sold artifact (`claude-opus-4-8`; switchable to Sonnet/Haiku for cost at scale).
FastAPI + Leaflet	Map dashboard, dossier download, before/after imagery, metrics.

The agent layer (value capture). `dossier_agent` (1) retrieves the most relevant statute chunks from pgvector, (2) drafts a **typed, fully-grounded** dossier with Claude (`messages.parse` → schema), and (3) runs an **adversarial critic** pass that rejects any claim not traceable to a detection fact or retrieved statute, revising once. Without an API key it emits a clearly-labelled rule-based template so the pipeline always completes.

4. Implementation and Evaluation

Change detection. The default detector is **spectral**: between two dates it flags pixels where the built-up index rises (ΔNDBI $\geq$ 0.06) and the vegetation index falls (ΔNDVI $\geq$ 0.12) — the signature of construction replacing crops — then vectorises connected regions $\geq$ 400 m^2 into candidate polygons with a confidence score. Requiring **both** signals suppresses the bare-soil / water false positives a single-index threshold would trip on (verified on a real 0%-cloud Changhua scene: NDVI ranged −0.21 to 0.62). The interface is swappable: a `torchgeo` Siamese-UNet (OSCD) path is a documented extension that falls back to spectral with a warning.

Zoning join — "illegal" = change + zoning. A change polygon is only a candidate violation once it sits on agricultural land. `zoning_join` intersects detections with farmland parcels in PostGIS, keeps only farmland overlaps, attaches the 地號, finds the nearest Disfactory label, and ranks by confidence. We never label a parcel "illegal" without the zoning overlay.

Evaluation — the credibility anchor. `evaluate` computes precision / recall of flagged detections against Disfactory labels within a match radius (default 80 m). Disfactory labels are crowd-reported and **incomplete**, so these are indicative, not absolute — a flagged parcel with no nearby label may be a real but unreported factory (precisely the value: surfacing the unreported). A verified offline demo run (synthetic bi-temporal imagery with injected building patches, deterministic sample labels) produced:

detections	flagged	true pos.	labels	matched	**precision**	**recall**	**F1**	radius
6	6	4	5	4	**0.67**	**0.80**	**0.73**	80 m

The score is deliberately non-trivial (two detected patches have no label → false positives; one labelled factory has no detectable patch → a miss), so the demo reports an honest figure rather than a staged 100%. The same metric is shown live on the dashboard for any run; the live pipeline runs the identical stages on a real Sentinel-2 pair for the AOI.

5. Scalability and Cost (10× / 100×)

Concern	Pilot (1 township)	10× (1 county)	100× (national)
Imagery	windowed COG reads, MinIO	partition by tile/date	real S3; Sentinel-1 SAR fusion to beat cloud
Compute	single-proc / local Spark	Spark, few executors	Spark cluster; CD is embarrassingly parallel
Store	one PostGIS	partition by AOI; read replicas	sharded PostGIS + parquet for cold detections
Agent	per-parcel Claude calls	prompt-cache statutes; Batch API	Batch + Haiku triage / Opus confirm; cents/parcel
Stream	one Kafka broker	partitions per county	multi-broker; consumer groups per region

Unit economics. A dossier is a few thousand tokens of context + output ≈ a few US cents per parcel, versus monitoring contracts in the hundreds of thousands of NT$ and volunteer-hours per detection. The compounding moat is the **imagery archive**: every cycle of snapshots is history a later competitor cannot reproduce.

6. Go-to-Market Difficulties (bonus)

- **Trust / adoption.** Answer with the precision / recall against known factories, a per-flag confidence score, and a **human-in-the-loop review** before any dossier is filed. The product is decision-support, never an automated ruling.
- **Data acquisition cost & licensing.** Sentinel-2 and Disfactory are open (CC BY 4.0 cited); statutes are MOJ open data. The real friction is the **NLSC vector cadastre** (地籍 WFS), which requires a government / academic application — so the demo ships a deterministic synthetic parcel layer (labelled `source='synthetic'`) and the production path joins the official farmland layer once access is granted. That gating becomes a moat once we are inside the channel.
- **Legal / ethics / PDPA.** The dossier targets parcels / structures, not persons; no private individuals are named; cautious "suspected / candidate" language.
- **Don't antagonise the incumbent.** We augment CET / Disfactory's proven workflow and the government's authority — leverage, not replacement.
- **Cold start.** Beachhead through the NGO channel; one county pilot before island scale; the eval number is the trust-builder that unlocks the first paid pilot.
- **Competition & moat.** A scraper of Disfactory is worthless (the data is free); the moat is the fused per-parcel substrate, the inspector-judgment scaffolding encoded in the detection→zoning→dossier pipeline, the compounding archive, and distribution inside the enforcement channel.

7. Caveats and Honesty

Cloud cover — Taiwan is cloudy; mitigated with low-cloud scene selection and multi-temporal windows; SAR fusion is a noted scale TODO. **Resolution** — 10 m is coarse for small buildings, used as a "something changed" trigger to be confirmed with higher-res orthophotos. **Eval honesty** — Disfactory labels are incomplete, so precision / recall are indicative; the synthetic parcel / imagery fallback is for offline reproducibility and is always labelled as such, never presented as real.

8. Reproducibility and Deliverables

One-command run: `make demo` (live) / `make demo-offline` (offline). Demand evidence: `scripts/demand/*` regenerate § 2's numbers. Tests: `make test` (unit + DB-integration, incl. zoning-join correctness). The repository contains all ingestion / processing / agent / delivery code, a runnable `README.md`, and an architecture overview mirroring § 3.

Criterion	**Weight**	**Where**
Target customer clarity	20%	§ 1
Demand evidence & acquisition	25%	§ 2 + `scripts/demand/`
Technical system & implementation	40%	§ 3– § 5 + the repository
Writing, diagrams, presentation	15%	this report + README diagram + dashboard
Bonus: GTM difficulties	+10%	§ 6
Bonus: live deployment	+10%	live URL on page 1 (optional)